

FOREST VERIFICATION REPORT

ANALYZED ZONE INFORMATION

Latitude	-1.7754	Longitude	29.4828
Zone size (km)	10	Analysis date	2026-09-30

61

FOREST INTEGRITY SCORE / 100 Degraded Forest

DETAILED RESULTS

Indicator	Value	Interpretation
Current vegetation (NDVI)	0.563	Good
NDVI Trend 3 years (Sentinel-2)	-0.018	Stable
10-year trend (Landsat 8/9)	+0.243	Improving
40-year trend (Landsat 1984)	+0.267	Improving
Forest Status (FAO)	Forest recovery	Recuperation
Annual NDVI rate	+0.67% / year	Amelioration
NDVI change 5 years	-0.065	Degrading
Leaf type	Broadleaf (Deciduous/Evergreen)	Flat leaves, high photosynthesis
Developmental stage	Growing forest (20-60 years)	Active growth phase
Above-ground biomass	~31.4 t/ha	Low biomass (<50 t/ha)
Canopy Height (GEDI+S2 10m)	6.8m max / 3.4m median	Tree height from LiDAR (ISS) fused with Sentinel-2 at 10m resolution
Carbon Assessment (IPCC)	536,428 t CO ₂	CO ₂ sequestered in this forest zone — IPCC methodology (AGB+BGB × 0.47 × 3.67)
CO ₂ per hectare	68.3 t CO ₂ /ha	Carbon density per hectare
Zone area	7,854 ha	Analysis zone area in hectares
Carbon credit value	EUR 8,046,420	Estimated value at EUR 15/tonne CO ₂ (voluntary carbon market)
REDD+ MRV Monitoring	Compliant	This report provides satellite-based Measurement, Reporting and Verification (MRV) data for the analyzed forest zone, consistent with REDD+ monitoring requirements. CO ₂ sequestration calculated using IPCC Tier 1 methodology from NASA GEDI LiDAR biomass data.

Methodology	IPCC 2006/2019 — Tier 1	Above-ground biomass measured via NASA GEDI LiDAR (25m, ISS). Carbon fraction: 0.47. CO2 conversion factor: 3.67. Below-ground biomass ratio: 0.26 (IPCC root-to-shoot). Source: IPCC Guidelines for National GHG Inventories (2019).
Carbon credit potential	EUR 670,535 / year	Estimated annual value based on maintained forest preservation. For official REDD+ credit issuance under VCS or Gold Standard, independent third-party verification is required.
GEDi Tree Cover	32.5%	Percentage of the zone covered by trees according to NASA GEDI
AI Forest Type	Savanna	AI-based classification
Deforestation Risk (AI)	Moderate	Moderate risk
Main cause (AI)	Climate-driven aridity combined with potential anthropogenic pressure in East African savanna region	
Degradation signs (AI)	Moderate NDVI (0.579) indicates sparse vegetation; negative NDWI (-0.581) suggests water stress and dry conditions typical of savanna ecosystems under pressure	
Recovery signs (AI)	EVI value of 1.603 indicates some photosynthetic activity; NBR (0.385) suggests no recent fire damage	
AI Recommendation	Monitor vegetation trends quarterly and implement water management strategies to enhance ecosystem resilience in this water-stressed savanna environment.	
Sentinel-1 Radar (VV)	-8.863 dB	Dense forest (radar)
Forest cover	88.9%	Excellent
Fire pixels detected (FIRMS/year)	13046 pixels	~3262 events estimated
Overall trend	stable	
Satellites used	- ESA Sentinel-2 - NASA FIRMS - ESA Sentinel-1 - NASA Landsat 8/9 - NASA Landsat 4/5 - NASA GEDI LiDAR	
Deforestation Alert	CAUTION — Light forest degradation	
Area change (3 years)	-330.3 ha	-3.59%
Annual rate (area/yr)	-110.1 ha/yr	Loss
Area change (10 years)	+7661 ha	+638.29%

Annual rate 10yr (area)	+766.1 ha/yr	Recovery
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SCORE BREAKDOWN

Component	Points	Maximum
Current vegetation (NDVI)	18	25
NDVI trend (3 years)	8	25
Forest cover	25	25
Sentinel-1 Radar (VV)	10	15
Fire score (0=many fires, 10=none)	0	10
TOTAL	61	100

FIRE DETECTION METHODOLOGY: Fire pixels are detected by NASA FIRMS using VIIRS (375m) and MODIS (1km) thermal anomaly sensors. Raw pixel count represents thermal detections over 12 months in the analysis zone. Estimated fire events are calculated using NASA clustering methodology (~4 pixels per fire event average). This is a statistical approximation. For precise fire event mapping, consult NASA FIRMS directly at firms.modaps.eosdis.nasa.gov

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